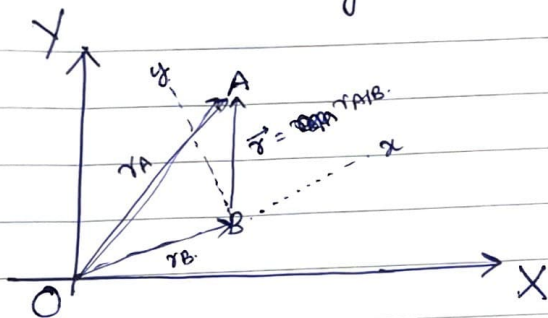


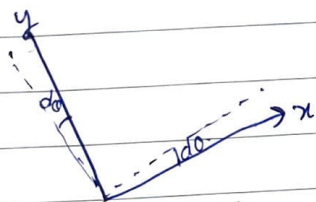
relative velocity w.r.t IAOR.



$(x, y) \rightarrow x-y$ axis.

$$r_A = r_B + \vec{r} = r_B + r_{B/A} = r_B + x\hat{i} + y\hat{j}$$

the $x-y$ coordinate axis is rotating so $x\hat{i}, y\hat{j}$ change with time and their derivatives are calculated below.



$$di = d\theta \hat{j}$$

$$\frac{di}{dt} = \frac{d\theta}{dt} \hat{j}$$

$$= \omega \hat{j}$$

$$(\dot{\hat{i}}) = \omega \times \hat{i} \quad \checkmark$$

$$dj = -d\theta \hat{i}$$

$$\frac{dj}{dt} = -\frac{d\theta}{dt} \hat{i}$$

$$(\dot{\hat{j}}) = \omega \times \hat{j} \quad \checkmark$$

To study motion of A w.r.t X-Y axis we need to take time derivative of r_A .

$$\begin{aligned} \dot{r}_A &= \dot{r}_B + \dot{x}\hat{i} + \dot{y}\hat{j} + x\dot{\hat{i}} + y\dot{\hat{j}} \\ &= \dot{r}_B + \omega \times (x\hat{i} + y\hat{j}) + \dot{r}_{B/A} \end{aligned}$$

$$v_A = v_B + \omega \times r_{B/A} + v_{B/A}$$

$$= \boxed{v_B + v_{B/A} + \omega \times r_{B/A}}$$